The Simplex method

- The graphical method can be applied to solve LP models involving only two decision variables.
- Simplex method is the method to solve (LPP) models which contain two or more decision variables.

- The simplex technique involves generating a series of solutions in tabular form.
- By inspecting the bottom (Index) row of each tableau, we can immediately determine if it represents the optimal solution.
- Each tableau corresponds to a corner point of the feasible space.

- The first tableau represents the Initial Basic Feasible Solution (IBFS), may correspond to the origin.
- Subsequent tableaus are developed by shifting to an adjacent corner point in the direction that yields the highest (smallest) rate of profit (cost).
- This process continues as long as a positive (negative) rate of profit (cost) exists.

The linear programming model can be written as:

$Max\ or\ Min$ Z$= \sum_{i=1}^{n} c_i x_i$

Subject to: $\sum_{i=1}^{n} a_{ij} x_j$; ∀ i = 1, 2, … , m

$x_i \geq 0$; ∀ i = 1, 2, … , n

Initialization (Standard form):

All variables should be non-negative Any unrestricted variable can be expressed as a difference of two non-negative variables

The RHS of each constraint should be non-negative If the RHS is negative, multiply both sides by (-1)

All constraints are express as equations Transform all the constraints to equality by introducing slack or surplus variables as follows:

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Non-Basic variables: Are the variables which coefficients are taking any of the values, whether positive or negative or zero. Slack, surplus & artificial variables:

a) If the inequality be $\leq$, then we add a slack variable + S(or S_n or x_{n+1} or p) to change to =.

$a_{11}x_1 + a_{12}x_2 + \ldots. + a_{1n}x_n \leq b_1$

$a_{11}x_1 + a_{12}x_2 + \ldots. + a_{1n}x_n + x_{n+1} = b_1$

Objective function

$Z = c_1x_1 + c_2x_2 + \ldots. + c_nx_n + 0x_{n+1}$

a) If the inequality be $\geq$, then we subtract a surplus variable - S (or S_n or x_{n+1} or p) to change to =

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \geq b_1$$

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n - x_{n+1} = b_1$$

Objective function

$$Z = c_1x_1 + c_2x_2 + \dots + c_nx_n - 0x_{n+1}$$

Cost of minimization (Convert the problem in to maximization)

Minimize Z= Maximize (-Z)

Case of maximization

Examine $Z_j - c_j$'s. If all are ≥ 0, stop the operation optimum solution is reached.

Then consider that x_j for which $Z_j - c_j$ is the least.

If $Z_j - c_j = 0$ all basic and non basic variable, then multiple solution exist from the LPP.

Consider the ratio $\left\{\frac{x_B}{y_{ij}}, y_{ij} > 0\right\}$.If all the $y_{ij} \leq 0$,stop the iteration. LPP has unbounderd solution

Solve the following LPP by simplex method

Max$Z = 2x_1 + 4x_2$

Subject to

$$2x_1 + 3x_2 \leq 48$$
$$x_1 + 3x_2 \leq 42$$
$$x_1 + x_2 \leq 21$$

And $x_1, x_2 \geq 0$

Solution

Let x_3, x_4 and x_5 are slack variables with cost coefficient '0'.

$$Z = 2x_1 + 4x_2 + 0x_3 + 0x_4 + 0x_5$$

$$2x_1 + 3x_2 + x_3 = 48$$
$$x_1 + 3x_2 + x_4 = 42$$
$$x_1 + x_2 + x_5 = 21$$

But our solution is

$$Z = 2x_1 + 4x_2 + 0x_3 + 0x_4 + 0x_5$$

$$2x_1 + 3x_2 + x_3 + 0x_4 + 0x_5 = 48$$

$$x_1 + 3x_2 + 0x_3 + x_4 + 0x_5 = 42$$

$$x_1 + x_2 + 0x_3 + 0x_4 + x_5 = 21$$

Identify initial basic feasible solution $x_1, x_2 = 0$

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$$Z = 2x_1 + 4x_2 + 0x_3 + 0x_4 + 0x_5$$

$$2x_1 + 3x_2 + x_3 + 0x_4 + 0x_5 = 48$$

$$x_1 + 3x_2 + 0x_3 + x_4 + 0x_5 = 42$$

$$x_1 + x_2 + 0x_3 + 0x_4 + x_5 = 21$$

Identify initial basic feasible solution $x_1, x_2 = 0,$ so, $x_3 = 48, x_4 = 42$ *and* x_5=21

Table 1

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	$X_B or b_i$
0	x_3	2	3	1	0	0	48
0	x_4	1	3	0	1	0	42
0	x_5	1	1	0	0	1	21
	Z_j	0	0	0	0	0	
	Z_j-C_j	-2	-4	0	0	0	Z=0

Calculus values of Z

$$0 \times 2 + 0 \times 1 + 0 \times 1 = 0$$
$$0 \times 3 + 0 \times 3 + 0 \times 1 = 0$$
$$0 \times 1 + 0 \times 0 + 0 \times 0 = 0$$
$$0 \times 0 + 0 \times 1 + 0 \times 0 = 0$$
$$0 \times 0 + 0 \times 0 + 0 \times 1 = 0$$

Next find the key (or pivot) column

If all Z_j-C_j are +ve then stop. Current solution is optimal.
If not chose the column with least -ve value ,
that is minimum Z_j-C_j.
Chose the entering variable
Min.$\{-2, -4\} = -4$
Therefore x_2 entering the basis

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
0	x_3	2	3	1	0	0	48
0	x_4	1	3	0	1	0	42
0	x_5	1	1	0	0	1	21
	Z_j	0	0	0	0	0	
	Z_j-C_j	-2	-4	0	0	0	Z=0

Calculus values of Z

$$0 \times 2 + 0 \times 1 + 0 \times 1 - 2 = -2$$
$$0 \times 3 + 0 \times 3 + 0 \times 1 - 4 = -4$$
$$0 \times 1 + 0 \times 0 + 0 \times 0 - 0 = 0$$
$$0 \times 0 + 0 \times 1 + 0 \times 0 - 0 = 0$$
$$0 \times 0 + 0 \times 0 + 0 \times -0 = 0$$

Therefore x_2 entering the basis

Next find the key row

Determine the ratio $\frac{x_B}{y_{ij}}$, for $y_{ij} > 0$

For the column corresponding to entering variable and take

Variable corresponding to minimum $\frac{x_B}{y_{ij}}$

Now minimum $\left\{\frac{48}{3}, \frac{42}{3}, \frac{21}{1}\right\} = 14$

Key raw is 2^{nd} row

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
0	x_3	2	3	1	0	0	48
0	x_2	1	3	0	1	0	42
0	x_5	1	1	0	0	1	21
	Z_j	0	0	0	0	0	
	Z_j-C_j	-2	-4	0	0	0	Z=0

Key (pivot) raw

Pivot element

Key (pivot)Colum

Next make key element =1, and other key elts should be make zero

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
0	x_3	2	3	1	0	0	48
4	x_2	1	3	0	1	0	42
0	x_5	1	1	0	0	1	21
	Z_j	0	0	0	0	0	
	Z_j-C_j	-2	-4	0	0	0	Z=0

Next make key element =1, and other key elts should be make zero

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
0	x_3	2	3	1	0	0	48
4	x_2	1	3	0	1	0	42
0	x_5	1	1	0	0	1	21
	Z_j	0	0	0	0	0	
	Z_j-C_j	-2	-4	0	0	0	Z=0

$$R_1' = R_1 - R_2$$

$$R_2' = \frac{R_2}{3}$$

$$R_3' = R_3 - \frac{1}{3}R_2$$

Next make key element =1

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
0	x_3	1	0	1	-1	0	6
4	x_2	$\frac{1}{3}$	1	0	$\frac{1}{3}$	0	14
0	x_5	$\frac{2}{3}$	0	0	$-\frac{1}{3}$	1	7
	Z_j	0	0	0	0	0	
	Z_j-C_j	-2	-4	0	0	0	Z=0

Therefore x_2 entering the basis, x_4 is a leaving variable

Table 2

	c_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
0	x_3	1	0	1	-1	0	6
4	x_2	$\frac{1}{3}$	1	0	$\frac{1}{3}$	0	14
0	x_5	$-\frac{2}{3}$	0	0	$-\frac{1}{3}$	1	7
	Z_j	$\frac{4}{3}$	4	0	$\frac{4}{3}$	0	
	Z_j-C_j	-$\frac{2}{3}$	0	0	$\frac{4}{3}$	0	Z=0

Calculus values of new Z

$$0 \times 2 + 4 \times \frac{1}{3} + 0 \times 1 = \frac{4}{3}$$
$$0 \times 0 + 4 \times 1 + 0 \times 0 = 4$$
$$0 \times 1 + 0 \times 0 + 0 \times 0 = 0$$
$$0 \times 0 + 4 \times \frac{1}{3} + 0 \times 0 = \frac{4}{3}$$
$$0 \times 0 + 4 \times 0 + 0 \times 1 = 0$$

Now minimum $\left\{-\frac{2}{3}, \frac{4}{3}\right\} = -\frac{2}{3}$

Therefore x_1 entering the basis

Identify the new key row and column

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
0	x_3	1	0	1	-1	0	6
4	x_2	$\frac{1}{3}$		0	$\frac{1}{3}$	0	14
0	x_5	$\frac{2}{3}$	0	0	$-\frac{1}{3}$	1	7
	Z_j-C_j	$-\frac{2}{3}$	0	0	$\frac{4}{3}$	0	Z=0

Now minimum $\left\{-\frac{2}{3}, \frac{4}{3}\right\} = -\frac{2}{3}$

Therefore x_1 entering the basis

Now minimum $\left\{\frac{6}{2}, \frac{14}{1/3}, \frac{7}{2/3}\right\} = 6$

The entering variable is x_1 and x_3

Is leaving variable

Identify the new key row and column

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
2	x_3	1	0	1	-1	0	6
4	x_2	$\frac{1}{3}$	1	0	$\frac{1}{3}$	0	14
0	x_5	$\frac{2}{3}$	0	0	$-\frac{1}{3}$	1	7
	Z_j-C_j	$-\frac{2}{3}$	0	0	$\frac{4}{3}$	0	Z=0

$$R_1' = R_1$$

$$R_2' = R_2 - \frac{1}{3}R_1$$

$$R_3' = R_3 - \frac{2}{3}R_1$$

Identify the new key row and column

	c_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
2	x_1	1	0	1	-1	0	6
4	x_2	0	1	$-\frac{1}{3}$	$\frac{2}{3}$	0	12
0	x_5	0	0	$-\frac{2}{3}$	$\frac{1}{3}$	1	3
	Z_j-C_j	$-\frac{2}{3}$	0	0	$\frac{4}{3}$	0	

$$R_1{}' = R_1$$

$$R_2{}' = R_2 - \frac{1}{3}R_1$$

$$R_3{}' = R_3 - \frac{2}{3}R_1$$

Table 3

Therefore optimal solution reached.

$x_1=6, x_2 = 12, x_5 = 3, x_3 = 0, x_4 = 0$

And MaxZ=60

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
2	x_1	1	0	1	-1	0	6
4	x_2	0	1	$-\frac{1}{3}$	$\frac{2}{3}$	0	12
0	x_5	0	0	$-\frac{2}{3}$	$\frac{1}{3}$	1	3
	Z_j	2	4	$\frac{2}{3}$	$\frac{2}{3}$	0	
	Z_j-C_j	0	0	$\frac{2}{3}$	$\frac{2}{3}$	0	

Z_j-$C_j \geq 0$

Therefore optimal solution reached.

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
0	x_3	1	0	1	-1	0	6
0	x_4	1	3	0	1	0	$\frac{42}{3} = 14$
0	x_5	1	1	0	0	1	$\frac{21}{1} = 21$
	Z_j-C_j	-2	-4	0	0	0	Z=0

Key raw

Key Colum

Pivot element

Identify the new key row and column

	c_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
0	x_3	1	0	1	-1	0	6
4	x_2	$\frac{1}{3}$	1	0	$\frac{1}{3}$	0	14
0	x_5	1	0	0	0	1	7
	Z_j-C_j	$-\frac{2}{3}$	0	0	$\frac{4}{3}$	0	Z=0

Now minimum $\left\{-\frac{2}{3}, \frac{4}{3}\right\} = -\frac{2}{3}$

Therefore x_1 entering the basis

Identify the new key row and column

Now minimum $\left\{\frac{16}{2}, \frac{14}{1/3}, \frac{21}{1}\right\} = 14$

	C_j	2	4	0	0	0	
C_B	BV	x_1	x_2	x_3	x_4	x_5	X_B
0	x_3	2	0	1	0	0	6
4	x_2	$\frac{1}{3}$	1	0	$\frac{1}{3}$	0	14
0	x_5	1	0	0	0	1	7
	Z_j-C_j	$-\frac{2}{3}$	0	0	$\frac{4}{3}$	0	Z=0